Stacking reveals a universal baryon fountain

Averaging thousands of simulated galaxies exposes coherent gas cycling from disk to halo.

arXiv:2609.21374 — Stacking galaxy ecosystems to reveal baryon cycling patterns

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Background

Galaxies continuously exchange gas with their halos through fountain-like inflows and outflows. These cycles regulate star formation and connect galaxies to their cosmic environment.

> **Individual galaxy ecosystems are too diverse and dynamic for their underlying baryon-cycle pattern to be seen directly.**

What this paper is selling

The authors stack gas properties across many simulated galaxies to suppress noise and reveal a strikingly regular fountain. The resulting framework links simulations to observations and quantifies the cycle.

Key results to remember

- Stacked velocity fields show a smooth, coherent fountain pattern across the full ecosystem.
- The fountain's geometry and extent scale predictably with stellar mass, halo mass, and redshift.
- A simple parameterized model reproduces the stacked fields, enabling quantitative comparisons.

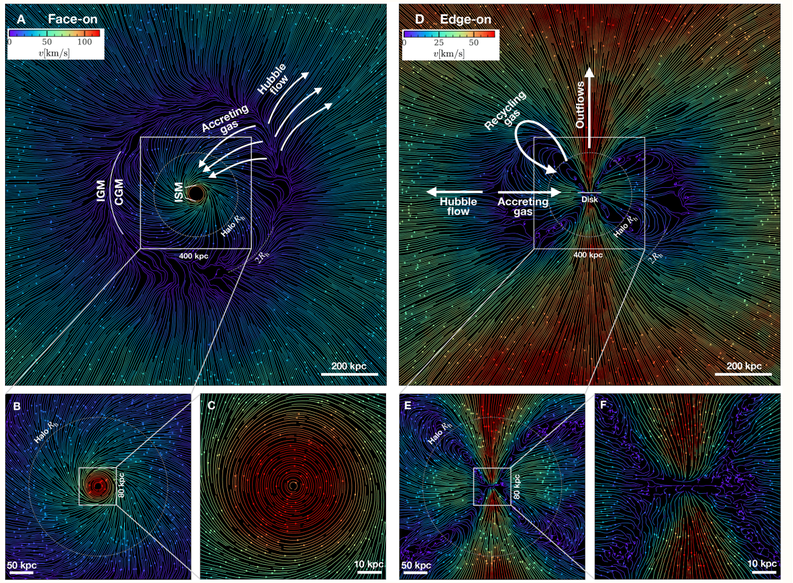


Figure 1 | Stacked velocity streamlines reveal a coherent, large-scale fountain pattern around an ensemble of Milky Way-like galaxies.

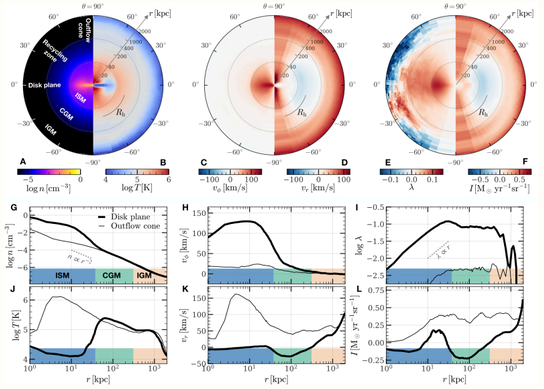


Figure 2 | Two-dimensional maps of stacked gas properties trace density, temperature, and radial flows across ISM, CGM, and IGM regimes.